

# Early Prediction of Characteristic Compressive Strength of Concrete Based on Mix Proportions Using Modified Dimensional Analysis

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**Abstract**—Concrete technology research has marked concrete's compressive strength as a critical indicator for predicting other vital mechanical properties. Existing models for forecasting this strength—from empirical formulas to complex machine learning algorithms—suffer from issues like lack of rationality and dimensional consistency. This study introduces the Modified Dimensional Analysis (MDA) method, which overcomes these challenges by ensuring dimensional homogeneity and mathematical stability, facilitating its use via simple tools like spreadsheets or calculators. We evaluated MDA against conventional models such as Abrams' law and various machine learning approaches, including decision trees, random forests, support vector machines, multiple linear regression, and artificial neural networks. Our findings reveal that MDA exhibits impressive data efficiency, requiring only 60% less training data than all other machine learning algorithms. It demonstrates a testing average R-square value of 0.94 and a maximum RMSE value of 5.55 MPa, which is competitive compared to models trained on 60% more datasets. Furthermore, if adjusted for specific construction sites, MDA will prove to be a robust on-field instrument, responsive to alterations in mix proportions, offering critical, timely insights for site engineers and supervisors. This adaptability allows for immediate, informed interventions, potentially improving construction outcomes.

**Keywords**—MDA, Concrete Compressive Strength, Construction Process Monitoring, Machine Learning, Dimensional Homogeneity

## I. INTRODUCTION

The compressive strength of concrete is considered one of the critical mechanical properties for determining the overall quality and compliance of concrete. Moreover, compressive strength is empirically co-related to other strength parameters, such as tensile strength and flexural strength, which are then utilized in the design of structures [1][2][3]. The characteristic compressive strength of concrete is determined by destructive test procedures on standard cubes or cylinders, typically at 28 days [4][5][53][54]. Before being placed, the concrete goes through an entire cycle. First, the mix design is finalized based on standard code and laboratory trials. Lastly, based on field conditions, necessary corrections are carried out in the mix design, and based on the results of field trials, the mix design is fine-tuned [4][6]. However, the final mix proportion often tends to change in the real-life application during the placement, and it becomes difficult to estimate the effect of such changes on the potential compressive strength of placed concrete and the sampled standard specimens thereof. Therefore, efforts have been made to estimate the compressive strength of concrete

reliably based on the actual mix proportion at placement. The Abrams law [1][7][8] though simple to use, considers only one parameter, i.e., the ratio of  $w/cm$  (water to cementitious material), and neglects the effects of other important parameters. While multiple regression model based on mix proportion [9][10] provides reliable quantitative results but is highly empirical and it lacks rational reasoning between independent and dependent parameters involved in the phenomena. Recently, many machine learning (ML) models [11][12][13][14] have been developed in this context, but these models require substantial data for the formulation and expert analytical skills in modeling the predictions [15][16][17], which limits the application of these ML models in the field.

Nonetheless, these models are highly empirical and lack rationality, dimensional homogeneity, and stability. Furthermore, only a few of the existing compressive strength prediction models have incorporated dimensional reduction in their formulation, which is helpful in practical problems arising due to an enormous group of variables in phenomena that lead to time consumption for determining correct relations [18]. However, none incorporates dimensional homogeneity, which is a significant requirement for a correct physical equation as stated by Fourier; otherwise, the equation becomes dimensionally invalid [19]. There is a need for an easy, reliable, and rational technique to predict the compressive strength of concrete that would incorporate real-time changes in mix proportion and help build the confidence of the concreting technician.

In this paper, the Modified Dimensional Analysis (MDA) method is used/employed to predict the mix-based compressive strength of concrete, which overcomes the limitations of dimensional homogeneity. This dimensional reduction makes the final formulation mathematical stable while keeping its application easy to use through a spreadsheet program or scientific calculator. The MDA method has been effectively used in other Civil engineering applications [20][21][22][23][24]. However, there is no literature on the application of modified dimensional analysis in predicting the mix-based compressive strength of concrete. This work uses a dataset of various concrete mix proportions from various sources to develop a generic methodology for early prediction of mix-based characteristics of compressive strength. The simple-to-derive and use formulation can be a field-handly tool. The key objective of this work is - to provide a rational, reliable estimation tool to be used in the field to predict the characteristics and compressive strength of concrete based on

mix proportion by considering the variations of mix proportions on field at the time of the placement.

The remainder of this work is structured as follows: Section A defines the research significance. Section II provides an overview of the dataset. Section III offers a brief description of the methodology. Section IV discusses the study's results and findings. Finally, Section V presents the conclusion, while Section VI details data availability.

#### A. Research Significance

This paper attempts to provide a generic and rational concrete mix proportion-based computational methodology that is easy to establish and use to predict the characteristic compressive strength of concrete by considering the variations of mix proportions on the field at the time of the placement. The multi-regression-based MDA model will help fine-tune and even optimize the laboratory and field concrete mix trials, potentially saving valuable resources. This methodology would also help build confidence in site engineers and concreting supervisors for concrete casting.

### II. DATASET

In this study, 650 dataset samples were used, which were taken from eleven different sources throughout the world from the laboratory as well as field [9][10][12][13][14][25][26][27][28][29][30]. These dataset samples were further sorted into six groups, i.e., Group A to Group F, depending upon the supplementary cementitious material used, i.e., fly ash, slag, and superplasticizer. For each group, computational models were formulated, and finally, for each group, all methods were compared to study the prediction efficiency of the models.

#### A. Group A: Water content, cement content, coarse aggregate, fine aggregates, fly ash, and Superplasticizers

As per the modified Buckingham-pi theorem [18], we have  $V = (\text{Water content, cement content, Coarse aggregate, fine aggregates, fly ash, Super plasticizers, grade, Compressive strength at 28 days})$  hence  $n=8$ . Representing  $V$  in dimensional form  $V = (M^1L^{-3}, M^1L^{-3}, M^1L^{-3}, M^1L^{-3}, M^1L^{-3}, M^1L^{-1}T^{-2}, M^1L^{-1}T^{-2})$ , where  $m=3$ . As a prerequisite the condition of  $D_{min}$ , which seeks to reduce the total number of primary dimensions in set  $V$ , is satisfied by minimizing the value of  $m$  and thus account for the greatest number of dimensionless groups possible. Therefore, assuming  $C = (M^1L^{-1}T^{-2})$ ,  $D = (M^1L^{-3})$ , hence,  $m=2$  (condition of  $D_{min}$ , is satisfied); number of repeating variables forming a set  $Q$ . Thus, number of dimensionless Pi-groups and also the number of isolated variables is  $N=(n-m)=8-2=6$ .  $R$  is the set of variables in  $V$  which have dimensions totally distinct from each other, therefore  $R = (\text{grade, water content})$ . Dimensionless groups are formed by the combination of the repeating ( $Q$ ) and isolated ( $NOTQ$ ) variable sets. Subsequently,  $Q$  is to be selected from  $R$ ; therefore,

$$Q = (\text{grade, cement content}) \quad (1)$$

, each of them occurs in each Pi group.

$$NOT-Q = (\text{cement content, coarse aggregate, fine aggregates, fly ash, Super plasticizer, Compressive Strength at 28 days}) \quad (2)$$

, each of them occurs only once in each Pi group. Therefore, the dimensionless groups are:

$$\pi_1 = (\text{grade, cement content, compressive strength}) \quad (3a)$$

$$\pi_2 = (\text{grade, cement content, water content}) \quad (3b)$$

$$\pi_3 = (\text{grade, cement content, coarse aggregate}) \quad (3c)$$

$$\pi_4 = (\text{grade, cement content, fine aggregate}) \quad (3d)$$

$$\pi_5 = (\text{grade, cement content, fly ash}) \quad (3e)$$

$$\pi_6 = (\text{grade, cement content, super plasticizer}) \quad (3f)$$

Hence, representing:

$$\pi_1 = (\text{Grade}^a, \text{water content}^b, \text{Compressive Strength}^c) \quad (4)$$

, where  $M^0L^0T^0 = (M^aL^{-a}T^{-2a})(M^bL^{-3b})(M^cL^{-c}T^{-2c})$ , comparing the indices of left hand side (LHS) and right hand side (RHS) we obtain  $M:0=a+b+c$ ,  $L:0=-a-3b-c$ , and  $T:0=-2a-2c$ . Hence,  $a = -c$  and  $b = 0$ , where in each case  $c=1$ . Substituting these values in (4), we obtain the Compressive strength/Grade in (5):

$$\pi_1 = (\text{Grade}^{-1}, \text{water content}^0, \text{Compressive Strength}^1) \quad (5)$$

Similarly, solving for  $(\pi_2, \pi_3, \pi_4, \pi_5, \pi_6)$ , we get:

$$\pi_2 = \text{Water/Cement} \quad (6)$$

$$\pi_3 = \text{Coarse aggregate/Cement} \quad (7)$$

$$\pi_4 = \text{Fine aggregate/Cement} \quad (8)$$

$$\pi_5 = \text{Fly-ash/Cement} \quad (9)$$

$$\pi_6 = \text{Super plasticizers/Cement} \quad (10)$$

Expressing  $\pi_1 = \varphi(\pi_2, \pi_3, \pi_4, \pi_5, \pi_6)$ , where  $\varphi$  is an unknown function and by substituting from (5) to (10), we have the Compressive-strength/Grade =  $\varphi(\text{Water/Cement, Coarse aggregate/Cement, Fine aggregate/Cement, Fly-ash/Cement, Super plasticizer/Cement})$ . To determine exact nature of  $\varphi$ , a power-product relationship of the dimensionless groups is used as follows:

$$\begin{aligned} \pi_1 &= \beta_1 \pi_2^{\beta_2} \pi_3^{\beta_3} \pi_4^{\beta_4} \pi_5^{\beta_5} \pi_6^{\beta_6}, \text{ or} \quad (11) \\ \left( \frac{\text{Compressive strength}}{\text{Grade}} \right) &= \beta_1 \left( \frac{\text{Water}}{\text{Cement}} \right)^{\beta_2} \left( \frac{\text{Coarse aggregate}}{\text{Cement}} \right)^{\beta_3} \dots \\ &\quad \left( \frac{\text{Fine aggregate}}{\text{Cement}} \right)^{\beta_4} \left( \frac{\text{Fly-ash}}{\text{Cement}} \right)^{\beta_5} \left( \frac{\text{Superplasticizer}}{\text{Cement}} \right)^{\beta_6} \\ \log \left( \frac{\text{Compressive strength}}{\text{Grade}} \right) &= \log(\beta_1) + \beta_2 \log \left( \frac{\text{Water}}{\text{Cement}} \right) \dots \\ &\quad + \beta_3 \log \left( \frac{\text{Coarse aggregate}}{\text{Cement}} \right) + \beta_4 \log \left( \frac{\text{Fine aggregate}}{\text{Cement}} \right) \dots \quad (12) \\ &\quad + \beta_5 \log \left( \frac{\text{Fly-ash}}{\text{Cement}} \right) + \beta_6 \log \left( \frac{\text{Superplasticizer}}{\text{Cement}} \right) \end{aligned}$$

Consequently, to solve (11), we convert it into its logarithmic form, where  $\pi_1$  is treated as the dependent variable and the value of constants  $(\beta_2, \beta_3, \beta_4, \beta_5, \beta_6)$  can be approximated by method of simultaneous solving of multi-variable equations or trial and error method. Now substituting six experimental results in (12) and generating system of equations to recognise value of constants  $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6$ .

$$\begin{aligned} 0.044427 &= \log(\beta_1) + 0.503905\beta_2 + 1.64333946\beta_3 \dots \quad (12a) \\ &\quad + 1.772026\beta_4 + 0.448216\beta_5 - 3.1849743\beta_6 \end{aligned}$$

$$0.086483 = \log(\beta_1) + 0.294239\beta_2 + 1.641803197\beta_3 \dots (12a) \\ + 1.769586\beta_4 + 0.44418\beta_5 - 3.2321211\beta_6$$

$$0.094271 = \log(\beta_1) + 0.35105\beta_2 + 1.079796017\beta_3 \dots (12c) \\ + 0.864997\beta_4 - 0.77539\beta_5 - 3.9252682\beta_6$$

$$0.123633 = \log(\beta_1) + 0.267879\beta_2 + 1.87931536\beta_3 \dots (12d) \\ + 1.49752\beta_4 + 0.446026\beta_5 - 3.2386785\beta_6$$

$$0.091262 = \log(\beta_1) - 0.61261\beta_2 + 1.081258274\beta_3 \dots (12e) \\ + 0.953027\beta_4 - 0.77373\beta_5 - 3.4339872\beta_6$$

$$0.033435 = \log(\beta_1) - 0.35208\beta_2 + 0.931655951\beta_3 \dots (12f) \\ + 0.953027\beta_4 - 0.77373\beta_5 - 3.4339872\beta_6$$

By solving the equations' system from (12a) to (12f):  $\beta_1 = 0.78$ ,  $\beta_2 = -0.18226$ ,  $\beta_3 = 0.069139$ ,  $\beta_4 = -0.05523$ ,  $\beta_5 = 0.152568$ ,  $\beta_6 = -0.09398$ , and by substituting above  $\beta$  values in (11), we obtain (13):

$$CS \text{ at 28 days} = (Grade)(0.78) \left[ \frac{water}{cement} \right]^{-0.18} \left[ \frac{Coarse Aggregates}{cement} \right]^{0.069} \dots \\ \left[ \frac{Fine Aggregates}{cement} \right]^{-0.055} \left[ \frac{Fly ash}{cement} \right]^{0.152} \left[ \frac{Super Platisizers}{cement} \right]^{-0.09} \quad (13)$$

#### B. Group B: Water content, cement content, coarse aggregate, fine aggregates, GGBS, and Superplasticizers

As a prerequisite from mix proportion 1 using modified dimensional analysis,  $V = (\text{Water content, cement content, coarse aggregate, fine aggregates, GGBS, Super plasticizers, grade, Compressive strength at 28 days})$ . Hence assuming,  $E = (M^1 L^{-1} T^{-2})$ ,  $F = (M^1 L^{-3})$ , therefore  $m=2$  (condition of  $D_{min}$ , is satisfied); number of repeating variables forming a set Q.  $N=(n-m)=8-2=6$ , being the number of dimensionless groups, and  $R=(\text{grade, water content})$ . Therefore, Q is considered the same as (1), and:

$$NOT-Q = (\text{cement content, coarse aggregate, fine aggregates, GGBS, Super plasticizer, Compressive Strength 28 days}) \quad (14)$$

Like mix proportion 1, a generic equation for mix proportion can be formulated in (15).

$$CS (28 \text{ days}) = (Grade)(1.128) \left[ \frac{water}{cement} \right]^{0.269} \left[ \frac{Coarse Aggregates}{cement} \right]^{-0.331} \dots \\ \left[ \frac{Fine Aggregates}{cement} \right]^{0.2384} \left[ \frac{GGBS}{cement} \right]^{-0.01} \left[ \frac{Super Platisizers}{cement} \right]^{-0.081} \quad (15)$$

#### C. Group C: Water content, cement content, coarse aggregate, fine aggregates, and Superplasticizers

As a prerequisite from mix proportion 1 and 2 using modified dimensional analysis,  $V = (\text{Water content, cement content, coarse aggregate, fine aggregates, Super plasticizers, grade, Compressive strength at 28 days})$ . Hence assuming,  $G = (M^1 L^{-1} T^{-2})$ ,  $H = (M^1 L^{-3})$ , therefore  $m=2$  (condition of  $D_{min}$ , is satisfied); number of repeating variables forming a set Q.  $N=(n-m)=7-2=5$  being the number of dimensionless groups, and  $R=(\text{grade, water content})$ . Therefore, Q is considered the same as (1), and:

$$NOT-Q = (\text{cement content, coarse aggregate, fine aggregates, Super plasticizer, Compressive Strength 28 days}) \quad (16)$$

Like mix proportion 1, 2, a generic equation for mix proportion can be formulated in (17).

$$CS (28 \text{ days}) = (Grade)(0.657) \left[ \frac{water}{cement} \right]^{0.05} \left[ \frac{Coarse Aggregates}{cement} \right]^{-0.00017} \dots \\ \left[ \frac{Fine Aggregates}{cement} \right]^{0.088} \left[ \frac{Super Platisizers}{cement} \right]^{-0.11} \quad (17)$$

#### D. Group D: Water content, cement content, coarse aggregate, fine aggregates, fly-ash, GGBS, and Superplasticizers

As a prerequisite from mix proportion 1, 2, and 3 using modified dimensional analysis,  $V = (\text{Water content, cement content, coarse aggregate, fine aggregates, fly-ash, GGBS, Super plasticizers, grade, Compressive strength at 28 days})$ . Hence assuming,  $I = (M^1 L^{-1} T^{-2})$ ,  $J = (M^1 L^{-3})$ , therefore  $m=2$  (condition of  $D_{min}$ , is satisfied); number of repeating variables forming a set Q.  $N=(n-m)=9-2=7$  being the number of dimensionless groups, and  $R=(\text{grade, water content})$ . Therefore, Q is considered the same as (1), and:

$$NOT-Q = (\text{cement content, coarse aggregate, fine aggregates, fly-ash, GGBS, Super plasticizer, CS 28 days}) \quad (18)$$

Like mix proportion 1, 2, and 3, a generic equation for mix proportion can be formulated in (19).

$$CS (28 \text{ days}) = (Grade)(1.36) \left[ \frac{water}{cement} \right]^{-0.04} \left[ \frac{Coarse Aggregates}{cement} \right]^{0.09} \dots \\ \left[ \frac{Fine Aggregates}{cement} \right]^{-0.21} \left[ \frac{Fly ash}{cement} \right]^{0.06} \left[ \frac{Slag}{cement} \right]^{-0.012} \left[ \frac{Super Platisizers}{cement} \right]^{0.0456} \quad (19)$$

#### E. Group E: Water content, cement content, coarse aggregate, and fine aggregates

As a prerequisite from mix proportion 1, 2, 3, and 4 using modified dimensional analysis,  $V = (\text{Water content, cement content, coarse aggregate, fine aggregates, Compressive strength at 28 days})$ . Hence assuming,  $I = (M^1 L^{-1} T^{-2})$ ,  $L = (M^1 L^{-3})$ , therefore  $m=2$  (condition of  $D_{min}$ , is satisfied); number of repeating variables forming a set Q.  $N=(n-m)=6-2=4$  being the number of dimensionless groups, and  $R=(\text{grade, water content})$ . Therefore, Q is considered the same as (1), and:

$$NOT-Q = (\text{cement content, coarse aggregate, fine aggregates, Compressive Strength at 28 days}) \quad (20)$$

Like mix proportion 1, 2, 3, and 4, a generic equation for mix proportion can be formulated in (21).

$$CS (28 \text{ days}) = (Grade)(0.7124) \left[ \frac{water}{cement} \right]^{-0.149} \dots \\ \left[ \frac{Coarse Aggregates}{cement} \right]^{0.0631} \left[ \frac{Fine Aggregates}{cement} \right]^{0.301} \quad (21)$$

#### F. Group F: Water content, cement content, coarse aggregate, fine aggregates, and GGBS

As a prerequisite from mix proportion 1, 2, 3, 4, and 5 using modified dimensional analysis,  $V = (\text{Water content, cement content, coarse aggregate, fine aggregates, GGBS, Compressive strength at 28 days})$ . Hence assuming,  $X = (M^1 L^{-1} T^{-2})$ ,  $Y = (M^1 L^{-3})$ , therefore  $m=2$  (condition of  $D_{min}$ , is satisfied); number of repeating variables forming a set Q.  $N=(n-m)=6-2=4$  being the number of dimensionless groups, and  $R=(\text{grade, water content})$ . Therefore, Q is considered the same as (1), and:

$$NOT-Q = (\text{cement content, coarse aggregate, fine aggregates, GGBS, CS at 28 days}) \quad (22)$$

Like mix proportion 1, 2, 3, 4, and 5, a generic equation for mix proportion can be formulated in (23).

$$CS (28 \text{ days}) = (Grade)(0.667) \left[ \frac{\text{water}}{\text{cement}} \right]^{0.088} \left[ \frac{\text{Coarse Aggregates}}{\text{cement}} \right]^{0.63} \dots \left[ \frac{\text{Fine Aggregates}}{\text{cement}} \right]^{-0.408} \left[ \frac{\text{GGBS}}{\text{cement}} \right]^{-0.077} \quad (23)$$

### III. METHODOLOGY

In this paper, an attempt has been made to formulate various prediction models using Abram's law, which is conventional law, multi-linear regression, four machine learning models, i.e., Decision Tree, Artificial Neural Network (ANN), Random Forest, and Support Vector Machine (SVM) and uniquely proposed MDA to estimate characteristic compressive strength based on mix proportion, and results from all these methods are compared.

In this study, 650 dataset samples were used, which were taken from eleven different sources throughout the world, from laboratories as well as fields. These dataset points were further sorted into six groups, i.e., Group A to Group F, depending upon the supplementary cementitious material used, i.e., fly ash, slag, and superplasticizer. For each group, the below-mentioned computational models were formulated, and finally, for each group, all methods were compared to study the prediction efficiency of the models.

The training of machine learning models, which include SVR, Decision tree, Random Forest, and ANN, is implemented in Python 3.6. The training processes are conducted in the Windows 10 platform with the processor of Intel Core i5 CPU 2.9 GHz and memory of 8 GB. The primary machine learning and Python libraries used in this work include scikit-learn 0.19.1 [31], NumPy 1.14.2 [32], SciPy 1.0.0 [33], pandas 0.23.1 [34], Keras [35], TensorFlow [36].

#### A. Abram's Law

Abram's law has been selected for comparison because of its academic popularity and behavior to yield (24). The law states that compressive strength is estimated through the water ratio to cementitious material for concrete mixture with workable consistency [1][7][8].

$$F_c = \left( \frac{A}{Bw/cm} \right) \quad (24)$$

, where A and B are empirical constants, and  $w/cm$  is water to cementitious ratio. For this methodology, the dataset in each group was distributed into formulation and testing datasets in the ratio 70:30. In this method, values of A and B are evaluated by converting the above equation into logarithmic form and then feeding 70% of data of each group to multi regression tool or similar regression tool for determining the value of coefficients which then are back substituted to estimate the value of A and B. The remaining 30% of data is tested with generated formulation for values of A and B, and the results are discussed in the next section. However the method is simple to use, but it fails in the case of similar water-to-cementitious material scenarios and does not incorporate all the mixed ingredients in the formulation, which does not provide reliable results.

#### B. Multi-linear Regression

Multi-linear regression has been extensively used in academics and industry to generate a linear relationship between the mix proportion used and the compressive strength of concrete to understand better the effect of mix ingredients on characteristic compressive strength [9][10]. The main objective of multi-linear regression is to assign the value of the coefficient and intercept it in a series of iterations such that the predicted value is the closest to the actual value, which is done by minimizing error or the cost function. In this paper, regression analysis is done through Excel to determine the general representation in (25).

$$y = b_0 + b_1x_1 + b_2x_2 + b_3 + b_4x_4 + b_5x_5 + b_nx_n \quad (25)$$

In this paper, the value of independent parameters was the Grade and mix of ingredients used, and the dependent parameter was characteristic compressive strength. The dataset was distributed into formulation and testing datasets in the ratio 70:30, using 70% of data for each group, and the value of constants and coefficients were estimated. Then, the formulation was generated through values of constants, and the intercept was tested on the remaining 30% of the dataset. However, if proper sampling and selection of independent parameters are not carried out in this method, then this method does not provide reliable results for the end users. Moreover, the relation between compressive strength and mixed ingredients is not always linear, and it fails to provide an equation that would provide a collective effect of all mixed ingredients used on the compressive strength of concrete [9].

#### C. Artificial Neural Network (ANN)

ANN was inspired by the biological workings of the Human Brain [37]. It consists of individual units called neurons responsible for learning. The ANN architecture has an input layer, a hidden layer, and an output layer. Each layer has activation functions such as Rectified Linear Unit (ReLU), tanh, sigmoid, linear, and other functions responsible for the architecture's non-linearity. The neurons in each layer pass on the information through the activation function, eventually to the output layer. Along with the information, some weight or coefficient also gets attached to it, which is done using backpropagation [14]. Initially, the weights are initialized randomly, but later, with training, the weights are optimized to achieve better results. The dataset used with ANN is divided into a 70:30 ratio for training and testing; further, 20% of the training data is used as a validation set.

The cost or loss function is used along with training to evaluate the trained model and optimize it. Particularly for regression, mean squared error is a standard loss function. An Optimization algorithm is used to minimize the loss function. Gradient descent is a common and popularly known optimizer. Thus, the lower the loss, the higher the accuracy. Therefore, training callbacks can stop the training if the validation loss increases after a certain threshold. Some parameters like learning rate, number of neurons, batch size, number of layers, and others are also responsible for optimizing the model; therefore, they must be initialized carefully and are known as hyperparameter tuning [38]. ANN model has higher generalization capability and has proven to be efficient for problems with high non-linearity [39]. However, the ANN

requires a lot of relevant data and takes a long time to train. Furthermore, ANN architecture is difficult to interpret and has a black-box effect [39][15]. In our study, the data is normalized, converting the dataset to the same scale, and boosting the model convergence. ANN has been implemented using Keras, TensorFlow, and Scikit Learn [31][40]. The ANN model is trained with an Input layer, one hidden layer, and one output layer, having a ReLU activation function and linear activation function at the output layer. Mean squared error is considered a loss, and for several epochs, callbacks have been used with validation loss as a threshold to stop training. K-fold cross-validation error is also monitored.

#### D. Decision Tree

Decision Tree algorithm is a popular supervised learning-based machine learning algorithm used for regression analysis [41]. A decision Tree constructs the regression model as a Tree diagram or flowchart using specific deterministic rules. Unlike ANN, the Decision Tree is a rule-based [42] model generated by logical splits in data and predicts the mapped target value. The algorithm aims to minimize the loss function or criteria such as mean squared error. The model is fit to independent variables with the target variable and is split at some points called leaf nodes in a dataset for each independent variable. The loss function or criteria is monitored, the split having minimum error is considered, and this process continues recursively. An efficient and optimized CART (Classification and Regression Tree) algorithm method is used in scikit learn to optimize the split. However, the Decision Tree can grow too large and overfit the data, causing less generalization on the test data and could inaccurately hinder the prediction. A threshold is put on several splits, or a minimum number of data points is considered for each leaf node to generate the Decision Tree to avoid a misclassification increase [43][12]. Our study uses Scikit learn packages to implement the Decision Tree algorithm [31][40]. Decision tree algorithms can be represented into quickly interpretable rules or logic statements. It does not need much training time compared to ANN and is flexible with different input responses [44][45]. However, it is more likely to overfit the data is unsteady and can sometimes be biased on some independent variables [44][45][46]. It does not provide a simple equation in the case of engineering applications.

#### E. Random Forest

A Random Forest is very similar to a Decision Tree, except a decision tree has a single tree, and a Random forest has multiple trees using an ensemble method [12]. A Bootstrap technique is applied to select each Decision Tree using different samples from the dataset:

- It implements the Bootstrap aggregation process, which will sample a similar dataset from the original dataset.
- Each dataset is used for generating individual trees in the ensemble.
- The randomness is added to decrease the correlation between each tree in the ensemble.

Different feature variables are used randomly for splitting the data to determine each node in each tree from the Random

Forest. The variable which minimizes the error for the target variable is selected for that node [47]. Similarly, each node and tree are determined, and a prediction is made. Each tree is responsible for its prediction, and then, finally, the results are averaged to output a single result. In our study, the Random forest algorithm is implemented using scikit learn Python packages [31][40]. This algorithm's prediction is based on multiple trees in the random forest, which is more generalized [38]. On the other hand, Random Forest is comparatively difficult to interpret and visualize compared to Decision Trees [47][38].

#### F. Support Vector Machine (SVM)

SVM is a supervised algorithm from the machine learning family invented by [48]. SVM is used for classification tasks, and a modified version called SVR (Support Vector Regression) is utilized to work with regression tasks. This modification creates a new higher dimension feature space linear model. The input features are mapped to this higher-dimension feature space, and an optimal separating hyperplane is determined. This can be done using kernel functions, which can be chosen before training. The margin boundary lines are then applied with some error threshold distance called - epsilon from the hyperplane. This loss is known as  $\epsilon$ -insensitive loss. The input points mapped out of the margin are not considered for training. The goal is to determine the hyperplane with the most deviation for the three actual target values optimally for all the samples in the training data [49][40]. Multiple Kernels are available, such as RBF (Radial Basis Function), poly (polynomial function), and linear function [51]. In our study, a linear function was implemented and achieved better results than other kernel functions. The SVR algorithm is implemented using Scikit learn packages, and hyperparameters have been tuned with manual guidance and GridSearch method [31][40][51]. SVM has a better generalization performance [52]. However, this algorithm has a high algorithmic complexity and requires time to optimize hyperparameters [52].

#### G. Modified Dimensional Analysis (MDA)

The primary purpose of modified dimensional analysis is to reduce dimensional space by introducing a set of secondary fundamental parameters in phenomena and, additionally, maintaining the dimensional homogeneity of the selected phenomena using the theorem of dimensional analysis. Considering phenomena consisting of  $n$  parameters involved, selected phenomena can be defined in set  $(V) = (V_1, V_2, V_3, \dots, V_n)$  and their dimensions are represented in terms of  $m$  independent primary dimensions  $(M, L, \text{ and } T)$ , i.e.,  $m=3$ . The value of  $m$  is redefined to 1 or 2 depending upon incorporating several secondary fundamental parameters, which tends to reduce the total number of primary dimensions in set  $V$  and thus account for the most significant number of dimensionless groups possible. Furthermore, dimensionless groups  $(\pi_1, \pi_2, \pi_3, \dots, \pi_N)$  can be arranged as a relation of dependent  $\pi$ -term to independent  $\pi$ -terms [18].

Expressing  $\pi_1 = \varphi(\pi_2, \pi_3, \pi_4, \pi_5, \pi_6)$ , where  $\varphi$  is an unknown function. To determine the exact nature of a power-product relationship of the dimensionless groups, it is used as follows in (26).

$$\pi_1 = \beta_1 \pi_2^{\beta_2} \pi_3^{\beta_3} \pi_4^{\beta_4} \pi_5^{\beta_5} \pi_6^{\beta_6} \quad (26)$$

, where  $\pi_1$  is treated as the dependent parameter and the value of constants ( $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6$ ) can be approximated by the method of simultaneous solving of multi-variable equations [24] or trial and error method [21] to formulate the final formulation for prediction of a dependent parameter. In this paper, the solving of multi-variable equations method was used for determining values of  $\beta$  for which selection of  $n$  number of random dataset points was done where  $n = \text{number of } \beta\text{-terms}$ . The final formulation matrix is presented in Fig. 1 to Fig. 6 for respective groups using MDA. The input parameters for each group's final formulation should be non-zero and non-negative.

| Total Parameters<br>in V. (n = 8)                                                                                                                                                                                                                                   | Dimensionless Pi Terms                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Water (kg/m <sup>3</sup> )<br>Cement (kg/m <sup>3</sup> )<br>Fly ash (kg/m <sup>3</sup> )<br>Coarse Aggregates (kg/m <sup>3</sup> )<br>Fine Aggregates (kg/m <sup>3</sup> )<br>Super Plasticizers (kg/m <sup>3</sup> )<br>Grade (MPa)<br>Compressive Strength (MPa) | N = n - m = 6<br>Water/Cement<br>Coarse Aggregates/Cement<br>Fine Aggregates/Cement<br>Super Plasticizers/Cement<br>Fly ash/Cement<br>Compressive Strength/Grade |
| <b>Primary Fundamental Terms</b><br>M = Mass<br>L = Length<br>T = Time                                                                                                                                                                                              | <b>Dataset Used for Formulation</b><br>6 (4.29 % of total dataset)                                                                                               |
| <b>Secondary Fundamentals Terms. (m = 2)</b><br>C = Grade = (M <sup>1</sup> L <sup>-1</sup> T <sup>-2</sup> )<br>D = Cement Content = (M <sup>1</sup> L <sup>-3</sup> )                                                                                             | <b>Dataset Used for Testing</b><br>134 (95.71 % of total dataset)                                                                                                |

Fig. 1. Formulation matrix for Group A using MDA

#### IV. RESULTS AND DISCUSSION

Results for Groups A to F were evaluated through modified dimensional analysis, Abram's law, multi-linear regression, and machine learning models – Decision tree, Random forest, SVM, and ANN. The results for all these models are discussed in this section. All the six equations formulated through MDA methodology were dimensionally homogenous, while no other method had a dimensional homogeneity aspect, which could be noticed in their result from (13), (15), (17), (19), (21), and (23). From Fig. 7, it can be seen that MDA requires the least amount of data in each group from A to F. Additionally, it is evident that MDA, on average, required 63% less data than any other methodologies.

Fig. 8 shows that MDA has tested, on average, 63% more data than other methodologies, along with maintaining a competitive average R-square value [54, 55, 56] and comparable maximum Root Mean Square Error (RMSE) [54, 55, 56] value concerning the machine learning and regression models. Abram's law yields the lowest R-square value and

highest RMSE value for each Group because it considers only the w/cm ratio and fails to capture the effects of other important parameters, which leads to unstable results. Moreover, Abram's law is not used for direct comparison with other methodologies but is included because of its exhaustive mention in past literature.

| Total Parameters<br>in V. (n = 8)                                                                                                                                                                                                                                | Dimensionless Pi Terms                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Water (kg/m <sup>3</sup> )<br>Cement (kg/m <sup>3</sup> )<br>Slag (kg/m <sup>3</sup> )<br>Coarse Aggregates (kg/m <sup>3</sup> )<br>Fine Aggregates (kg/m <sup>3</sup> )<br>Super Plasticizers (kg/m <sup>3</sup> )<br>Grade (MPa)<br>Compressive Strength (MPa) | N = n - m = 6<br>Water/Cement<br>Coarse Aggregates/Cement<br>Fine Aggregates/Cement<br>Super Plasticizers/Cement<br>Slag/Cement<br>Compressive Strength/Grade |
| <b>Primary Fundamental Terms</b><br>M = Mass<br>L = Length<br>T = Time                                                                                                                                                                                           | <b>Dataset Used for Formulation</b><br>6 (5.66 % of total dataset)                                                                                            |
| <b>Secondary Fundamentals Terms. (m = 2)</b><br>C = Grade = (M <sup>1</sup> L <sup>-1</sup> T <sup>-2</sup> )<br>D = Cement Content = (M <sup>1</sup> L <sup>-3</sup> )                                                                                          | <b>Dataset Used for Testing</b><br>100 (94.34 % of total dataset)                                                                                             |

Fig. 2. Formulation matrix for Group B using MDA

| Total Parameters<br>in V. (n = 7)                                                                                                                                                                                                   | Dimensionless Pi Terms                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Water (kg/m <sup>3</sup> )<br>Cement (kg/m <sup>3</sup> )<br>Coarse Aggregates (kg/m <sup>3</sup> )<br>Fine Aggregates (kg/m <sup>3</sup> )<br>Super Plasticizers (kg/m <sup>3</sup> )<br>Grade (MPa)<br>Compressive Strength (MPa) | N = n - m = 5<br>Water/Cement<br>Coarse Aggregates/Cement<br>Fine Aggregates/Cement<br>Super Plasticizers/Cement<br>Compressive Strength/Grade |
| <b>Primary Fundamental Terms</b><br>M = Mass<br>L = Length<br>T = Time                                                                                                                                                              | <b>Dataset Used for Formulation</b><br>5 (11.11 % of total dataset)                                                                            |
| <b>Secondary Fundamentals Terms. (m = 2)</b><br>C = Grade = (M <sup>1</sup> L <sup>-1</sup> T <sup>-2</sup> )<br>D = Cement Content = (M <sup>1</sup> L <sup>-3</sup> )                                                             | <b>Dataset Used for Testing</b><br>40 (88.89 % of total dataset)                                                                               |

Fig. 3. Formulation matrix for Group C using MDA

| Total Parameters<br>in V. (n = 9)                                                                                                                                                                                                                                                                | Dimensionless Pi Terms                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Water (kg/m <sup>3</sup> )<br>Cement (kg/m <sup>3</sup> )<br>Slag (kg/m <sup>3</sup> )<br>Fly ash (kg/m <sup>3</sup> )<br>Coarse Aggregates (kg/m <sup>3</sup> )<br>Fine Aggregates (kg/m <sup>3</sup> )<br>Super Plasticizers (kg/m <sup>3</sup> )<br>Grade (MPa)<br>Compressive Strength (MPa) | N = n - m = 7<br>Water/Cement<br>Coarse Aggregates/Cement<br>Fly ash/Cement<br>Fine Aggregates/Cement<br>Super Plasticizers/Cement<br>Slag/Cement<br>Compressive Strength/Grade |
| <b>Primary Fundamental Terms</b><br>M = Mass<br>L = Length<br>T = Time                                                                                                                                                                                                                           | <b>Dataset Used for Formulation</b><br>7 (3.87 % of total dataset)                                                                                                              |
| <b>Secondary Fundamentals Terms. (m = 2)</b><br>C = Grade = (M <sup>1</sup> L <sup>-1</sup> T <sup>-2</sup> )<br>D = Cement Content = (M <sup>1</sup> L <sup>-3</sup> )                                                                                                                          | <b>Dataset Used for Testing</b><br>174 (96.13 % of total dataset)                                                                                                               |

Fig. 4. Formulation matrix for Group D using MDA

| Total Parameters<br>in V. (n = 6)                                                                                                                                                        | Dimensionless Pi Terms                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Water (kg/m <sup>3</sup> )<br>Cement (kg/m <sup>3</sup> )<br>Coarse Aggregates (kg/m <sup>3</sup> )<br>Fine Aggregates (kg/m <sup>3</sup> )<br>Grade (MPa)<br>Compressive Strength (MPa) | N = n - m = 4<br>Water/Cement<br>Coarse Aggregates/Cement<br>Fine Aggregates/Cement<br>Compressive Strength/Grade |
| <b>Primary Fundamental Terms</b><br>M = Mass<br>L = Length<br>T = Time                                                                                                                   | <b>Dataset Used for Formulation</b><br>4 (3.22 % of total dataset)                                                |
| <b>Secondary Fundamentals Terms. (m = 2)</b><br>C = Grade = (M <sup>1</sup> L <sup>-1</sup> T <sup>-2</sup> )<br>D = Cement Content = (M <sup>1</sup> L <sup>-3</sup> )                  | <b>Dataset Used for Testing</b><br>120 (96.78 % of total dataset)                                                 |

Fig. 5. Formulation matrix for Group E using MDA

The R-square or coefficient of determination is a statistical parameter that indicates a correlation between the actual experimental value and the estimated value calculated through the chosen model. It has been mentioned that an R-square value above 0.8 for concrete compressive strength application is considered a good fit [13]; interestingly, Fig. 9 highlights that the R-square value of the MDA method used for all Groups is

above 0.9, which is much above the considered limit. The corresponding R-square value is comparable for all methodologies used in this research except for Abram's law for each group. However, it can be seen that for Group C, which had the least data compared to all Groups, MDA had an R-square value that was more significant than that of other machine learning methods for this group. This can be attributed to the behavior of machine learning methods requiring substantial data for modeling and the advantage of MDA requiring substantially fewer data, i.e., 60 percent fewer dataset points.

| Total Parameters<br>in V. (n = 7)                                                                                                                                                                                     | Dimensionless Pi Terms                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Water (kg/m <sup>3</sup> )<br>Cement (kg/m <sup>3</sup> )<br>Slag (kg/m <sup>3</sup> )<br>Coarse Aggregates (kg/m <sup>3</sup> )<br>Fine Aggregates (kg/m <sup>3</sup> )<br>Grade (MPa)<br>Compressive Strength (MPa) | N = n - m = 5<br>Water/Cement<br>Coarse Aggregates/Cement<br>Fine Aggregates/Cement<br>Slag/Cement<br>Compressive Strength/Grade |
| <b>Primary Fundamental Terms</b><br>M = Mass<br>L = Length<br>T = Time                                                                                                                                                | <b>Dataset Used for Formulation</b><br>5 (11.11 % of total dataset)                                                              |
| <b>Secondary Fundamentals Terms. (m = 2)</b><br>C = Grade = (M <sup>1</sup> L <sup>-1</sup> T <sup>-2</sup> )<br>D = Cement Content = (M <sup>1</sup> L <sup>-3</sup> )                                               | <b>Dataset Used for Testing</b><br>40 (88.88 % of total dataset)                                                                 |

Fig. 6. Formulation matrix for Group F using MDA

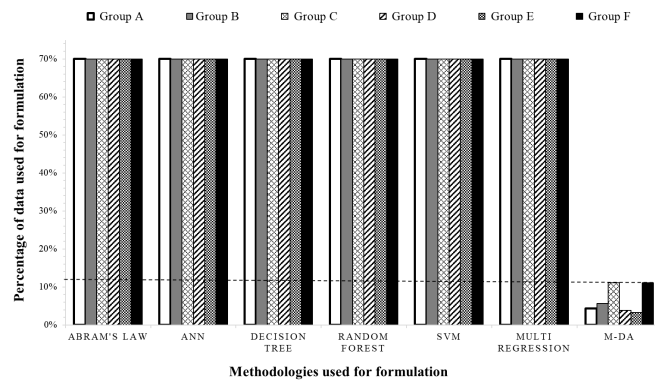


Fig. 7. Percentage of data used for formulation of various techniques.

The RMSE is a statistical parameter used to measure error for estimated models quantitatively. Considering Fig. 10, all methodologies used except Abrahm law for each group, the corresponding R-square value is comparable. Moreover, for Group C, which had the least dataset among all Groups, the value of RMSE for MDA for this Group is less than that of



machine learning models. This indicates MDA's reliability and stability for predicting compressive strength with a specific low error envelope. However, multi-regression had low RMSE, but it required 58% more data for formulation and tested 59% fewer dataset points. The coefficient of variation determined for RMSE and R square for MDA methodology was equal to 33% and 2%, reported least among all other methodologies. This determines MDA's reliable, stable, and robust approach to accurately predict concrete compressive strength with a specific low envelope of error and does not depend upon the number of dataset points in the analysis.

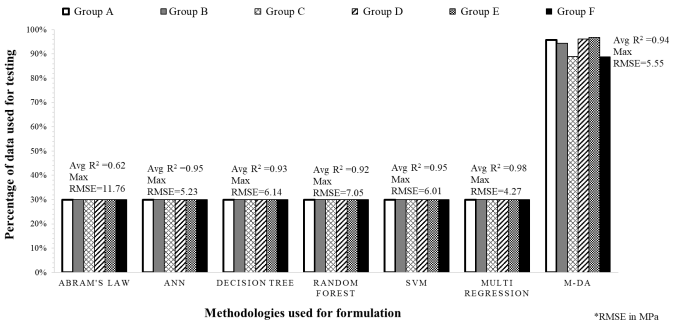


Fig. 8. Percentage of data used for testing of various techniques.

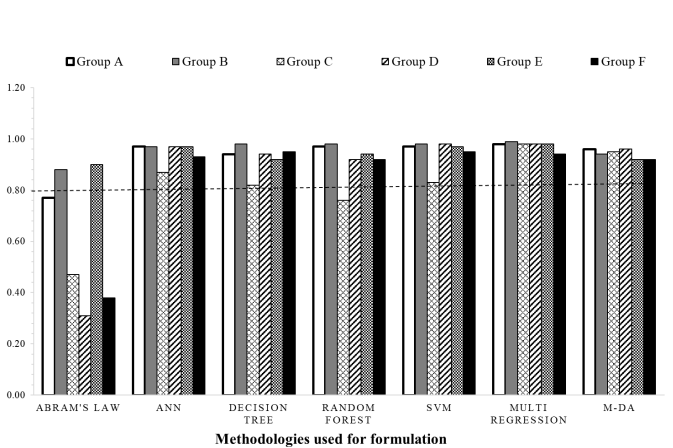


Fig. 9. R-square values from the different modeling techniques.

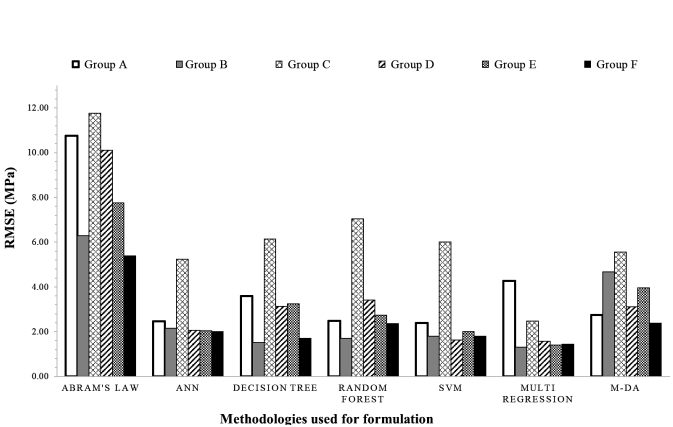


Fig. 10. RMSE (in MPa) from the different modeling techniques.

From Fig. 11, MDA has majorly predicted more conservative values than any other method. This determines the ability of the MDA method to predict the compressive strength of concrete in a given considerable safety limit for construction. Fig 12 shows that the standard deviation value for all the methodologies in each Group except Abrams law does not show a significant change.

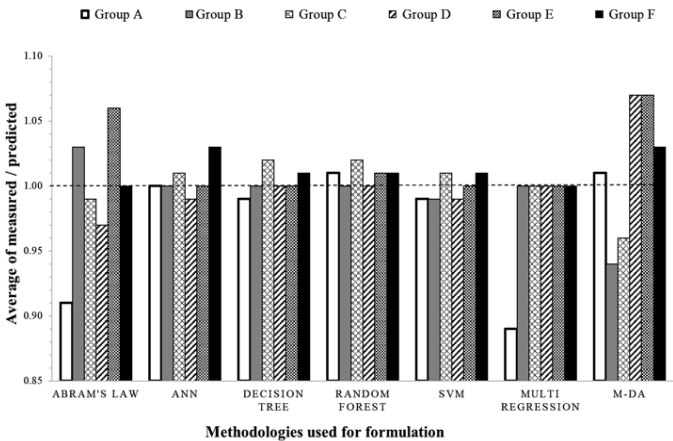


Fig. 11. Average of measured/predicted from all modeling techniques.

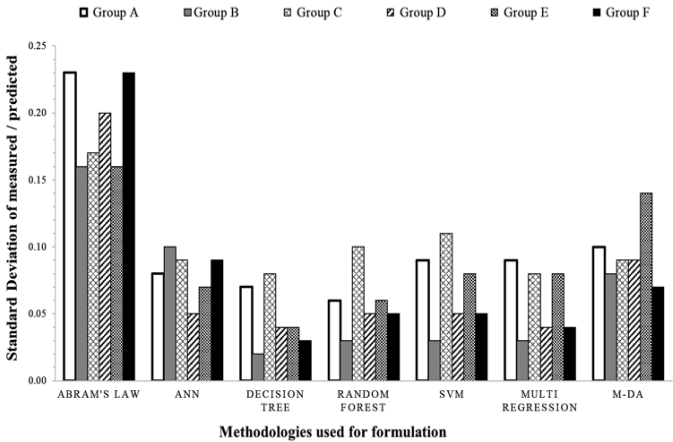


Fig. 12. Standard deviation of measured/predicted from all modeling techniques.

### V. CONCLUSIONS

The MDA methodology is a more reliable technique for the prediction of compressive strength of concrete as it provides comparable results and requires the least amount of data for formulation compared to any other methodology. The least coefficient of variation of RMSE and R-square for MDA compared to other methodologies reflects the mathematical stability of the MDA, which is an essential aspect of any mathematical methodology.

Group C results for R-square and RMSE show that MDA is independent concerning the number of dataset points used in the analysis. MDA does not require expert analytical skills; a person with essential analytical ability could apply it to concrete applications using a spreadsheet or scientific calculator. MDA has provided dimensional reduction, which solves the problem



of reducing total variables to the correct subset of variables, which helps save time for the analysis, and dimensional homogeneity, which helps select the correct correlation between dependent and independent parameters.

Results indicate that if MDA is fine-tuned according to site specification, it could be used as a field handy tool to capture real-time changes in mix proportion throughout the construction procedure. Furthermore, this tool would provide valuable insights to site engineers and concrete supervisors, which would help them take timely and effective mitigating measures in case of casualties.

## VI. DATA AVAILABILITY

All datasets, analytical models and codes generated during the study are available from the corresponding author (N. Pawar; nikhilpawar.nik433@gmail.com) by request.

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